

Report - Further Experiments with RAG

Fall 2026

In the second part of the lab, I implemented a RAG system using Qdrant together with SpeechState. The goal was to create a system that would be able to address questions and help old and new students studying at the university of Gothenburg. To achieve that, documents from the GU Student Portal were split into chunks, converted into embeddings, and finally stored in a Qdrant collection. These embeddings will be later used to retrieve the five most relevant chunks from the collection; the chunks will be then included in the system prompt before generating the response with the LLM.

At first glance, the basic system worked well for direct questions. When I asked, for example, the system “What is the phone number for Feelgood?”, it retrieved the relevant Student Healthcare document and answered with the correct phone number 031-786 43 22. However, when it comes to longer dialogues and questions unrelated to the university, I noticed some limitations that I wanted to further explore and discuss. Thus, I decided to focus mainly on these two questions: whether RAG works well when a dialogue unfolds rather than only for one-shot queries, and whether it is better to use only the latest user utterance when querying the database. I also experimented with the number of results returned from Qdrant.

The first thing that I noticed is that retrieval was performed for every recognized user utterance, even when no external information was required. For example, in the beginning of the conversation, when I said to the system “Hey there”, the system still tried to query Qdrant. However, there were no relevant documents that answer greeting expressions, so the system returned the most similar available chunks. Those chunks had to do with information about new students and university library services. The LLM then used this retrieved context and responded as if the user was asking about the university library. A similar situation occurred with utterances such as: “Thank you so much for that”. Retrieval was performed even though the utterance was conversational rather than informational. This limitation of the simple RAG architecture is very important, since a vector database will still return its nearest results even when none of them are actually relevant. Consequently, this irrelevant retrieval can influence the generated response from the LLM. A possible improvement would be to add a decision step before retrieval. Informational queries could then use Retrieval, while greetings, acknowledgements, and chit-chat could go directly to ChatCompletion.

During my first experiment, I tried to test whether using dialogue context could improve retrieval. Initially, Qdrant was queried using only the latest user utterance. This worked well for independent questions, but caused problems with context-dependent follow-up questions. When I first asked “What is the phone number for Feelgood?”, the system retrieved the correct information. When I continued by asking “And what is their address?”, the LLM was able to understand that “their” referred to Feelgood because it had access to the conversation history. However, the Retrieval actor only received the latest question, so Qdrant did not know what “their” referred to and returned unrelated information. That is why I decided to modify the system so that the retrieval would use the last two messages from the messages history. The retrieval query therefore contained both the original Feelgood question and the follow-up question. After repeating the experiment, Qdrant retrieved the relevant Student Healthcare information, including the address Östra Hamngatan 26, and the system answered correctly.

I also tested whether including a previous message could cause problems when the topic changed. I first asked about the Feelgood phone number and then I continued with a question like “how to contact Disability Study Support?”. Even though both questions were included in the retrieval query, Qdrant still retrieved the correct Disability Study Support information. This suggests that using recent dialogue context can help with ambiguous follow-ups without necessarily harming explicit questions. However, using the last two messages may not always work well, since the previous message could be unrelated to the current question.

During my second experiment, I investigated the number of results returned by Qdrant. The original system returned the five most similar chunks, so I compared this with returning only one result. I used the question “I need help with my studies because of a disability. Who should I contact and when can I call?”. With `limit:1`, the retrieved chunk already contained enough information for the system to provide the relevant email address and phone hours. I then repeated the same experiment with `limit:5`. The answer was also correct, but the retrieved context was much larger and contained additional information that was not needed for the question. For this example, retrieving five results did not noticeably improve the answer compared with retrieving one. However, this does not mean that one result is always preferable. More complicated questions may require information from several chunks. The number of results therefore represents a trade-off between providing enough information and introducing unnecessary context.

From these experiments, I noticed that adding more context does not always make the RAG system better. Using only the latest user message was not enough for some follow-up questions, while using the last two messages helped the system retrieve more relevant information. However, this could also become a problem if the previous message is about a different topic. I also found that using five retrieved results was not necessary for the question I tested, since one result was already enough to give a correct answer. Another issue was that the system performed retrieval even for simple messages like “Hello”, which sometimes added unrelated information to the response. In the future, the system could first decide if retrieval is actually needed and use different amounts of context depending on the question. Overall, I think these experiments showed that choosing the right context is an important part of making RAG work well in a dialogue system.